Methodology and presentation

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What is a research paper

It is a paper that

- Present an Original approach to some academic topic
- Organized around a central thesis
- Using insights based on research
- citing sources in scholarly fashion

Academic paper

Original Approach

Own idea

Primary and Secondary Sources

Background reading and look for research sources

- strengthen
- challenge
- modify
- illustrate

- Primary sources are “raw materials”

- Secondary sources

Your First Source: The Library

- Books
- Periodicals
- Other printed materials
- Non-printed materials
- Computer access to Internet sources

The World of Reference Materials

Here are nine main categories of reference materials

- Encyclopedias (general and specialized)
- Biographical dictionaries
- Periodical indexes
- Dictionaries of quotations and concordances
- Atlases and gazetteers
- Chronologies
- Dictionaries (general and specialized)
- Handbooks and sourcebooks
- Bibliographies

Scholars

Building new knowledge

- Universitiy
- research centers
- journalists

Practice of reading and writing

Why learning how to write a research paper

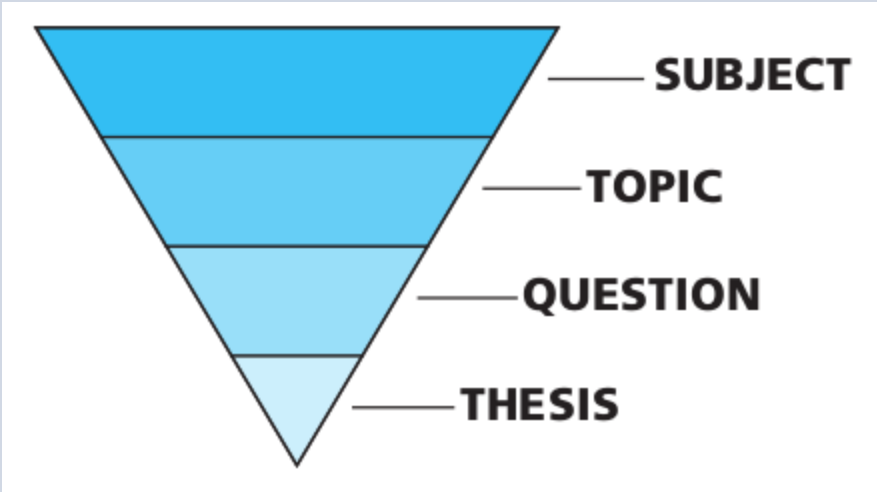
- draw information from several sources
- Practical life reasons
  - economics
  - any kind of decisions
  - Understanding how to
    - analyze
    - evaluate
    - criticize
    - combine
    - draw conclusions
  - Who is telling truth
  - Even buying a house can be research

The Definition of a Research Paper

A research paper is a paper that

- presents an original approach to some academic topic,
- organized around a central thesis,
- using insights based on research, and
- citing sources in a scholarly fashion.

The Hierarchy of a Research Paper



Subject

1. Look at the Table of Contents in the textbook

1. Take a stroll through the library or bookstore

1. Ask your teachers or professors for some guidance

Topic

- Think about
  1. an important turning point in the events within your subject.
  1. some fact you discovered that surprised you about your subject
  2. an oversimplification
  3. a trend or “story”
  4. two or more situations or personalities within your subject

1. Take a stroll through the library or bookstore
2. Ask your teachers or professors for some guidance

- Your topic shouldn’t be so
  1. broad that literally hundreds of books or articles

2. narrow that only a handful of books or articles have been written about it

A good topic

- is not merely a narrative
- is not too broad
- is not too narrow
- has research sources
- has not been overdone
- interests you

Question

Your research question is just the question or "puzzle" about the topic that your research paper will attempt to answer or solve

Thesis

A working thesis is a hypothetical answer to your research question — one that will guide you during your research. It probably won't be the exact thesis you'll discuss in your final research paper, but it might be.

A good working thesis

- can be stated in a single clear sentence,
- is a positive, non-obvious statement, and
- can be tested through research.

Bibliographical data to record for any sources that might potentially be helpful

- Call number
- Name of the author/editor
- Title and subtitle
- Edition/volume/issue numbers
- Place of publication
- Name of the publisher
- Copyright date
- Chapter/section/column/page numbers
- Features of the book that seem promising (especially bibliographies)

The Preliminary Bibliography

Your preliminary bibliography should include

- A few current and complete background books.
- The most important primary sources — described if not named.
- A few key secondary sources

Search Engines and Meta Search Engines

- Google
- presearch.org
- duckduckgo.com
- privado
- Yahoo
- Bing
- Ask.com
- AltaVista
- Gigablast
- Dogpile
- MetaCrawler

- filetype: pdf
- copyright:
- "Strict phrase"
- - does not occure
- regular expression
- ( ) + or and
- refine results based on date and other options
- site: yoosofan.github.io
-

Scientific Online Library

- https://scholar.google.com/
- https://www.ipl.org/
- https://www.answers.com/
- https://www.dictionary.com/
- https://www.thesaurus.com/
- https://www.wikipedia.org/
- https://news.google.com/
- https://doaj.org/
- https://refdesk.com/
- https://loc.gov/
- https://www.infoplease.com/
- http://lii.org/
- http://digital-librarian.com/
- https://archive.org/
- http://archive.is/
- https://www.researchgate.net
- https://www.academia.edu/
- https://archive-it.org/
- https://news.ycombinator.com/
- https://books.google.com
- https://citeseerx.ist.psu.edu/
- https://www.sciencedirect.com/

Article Samples

My articles

Article Formats

Persian Scientific Online Library

- https://www.sid.ir/
- https://ricest.ac.ir/
- https://en.civilica.com
- https://irandoc.ac.ir/

Bibliographic Tools

- Organise citation
  1. Zotero (https://www.zotero.org/)
  2. Mendely (https://www.mendeley.com/)
  3. https://endnote.com/
- Other Tools
  1. https://www.mybib.com/
  2. https://zbib.org/
  3. https://www.bibme.org
  4. https://www.citationmachine.net
  5. https://www.citefast.com
  6. https://docear.org/
  7. https://www.bibliography.com/

Forms of Scientific Publications

- Book
- Thesis
- Article

Book

- A form of publication
- Most people are familiar with
- Does not contain new results
- Does not provide evidence for the correctness of the information they present.
- Collect information and present it in
    - an accessible
    - readable form
- Textbooks are generally better written than papers.
- Established scientific knowledge

TextBook

- Mostly for undergraduate students
- However, there are excellent books for graduate students

Thesis

- Mostly University thesis
- deep or definitive exploration of a single problem
- More inclusive than papers may include
    - Uninteresting failures,
    - Invalid hypotheses,
    - Misconceptions, and
    - Experimental mistakes
- objective addition to scientific knowledge
- not a description of the path that was taken to the result
- Style is not just about how to write, but is also about what to say

Journal Paper

- End product of the research process
- A careful presentation of new ideas
- That has been revised according to suggestions or criticisms of
  - referees
  - colleagues
- Consists of
  - Arguments,
  - evidence,
  - experiments,
  - proofs, or
  - Background required to support and explain a central hypothesis
- content of a paper must be
  - defended and
  - justified

- The process of research that leads to a paper can include
  - Uninteresting failures,
  - Invalid hypotheses,
  - Misconceptions, and
  - Experimental mistakes
  - With few exceptions these do not belong in a paper.
- objective addition to scientific knowledge
- not a description of the path that was taken to the result
- Style is not just about how to write, but is also about what to say

A paper in conference proceedings

- End-product
- To report work in progress.
- Limited opportunities for
  - iteration
  - revision
- Maybe constrained by strict length limits

extended abstract

- Omitting detail of the work
- review the results of a research program
- may not include enough detail to make a solid argument for the claims.

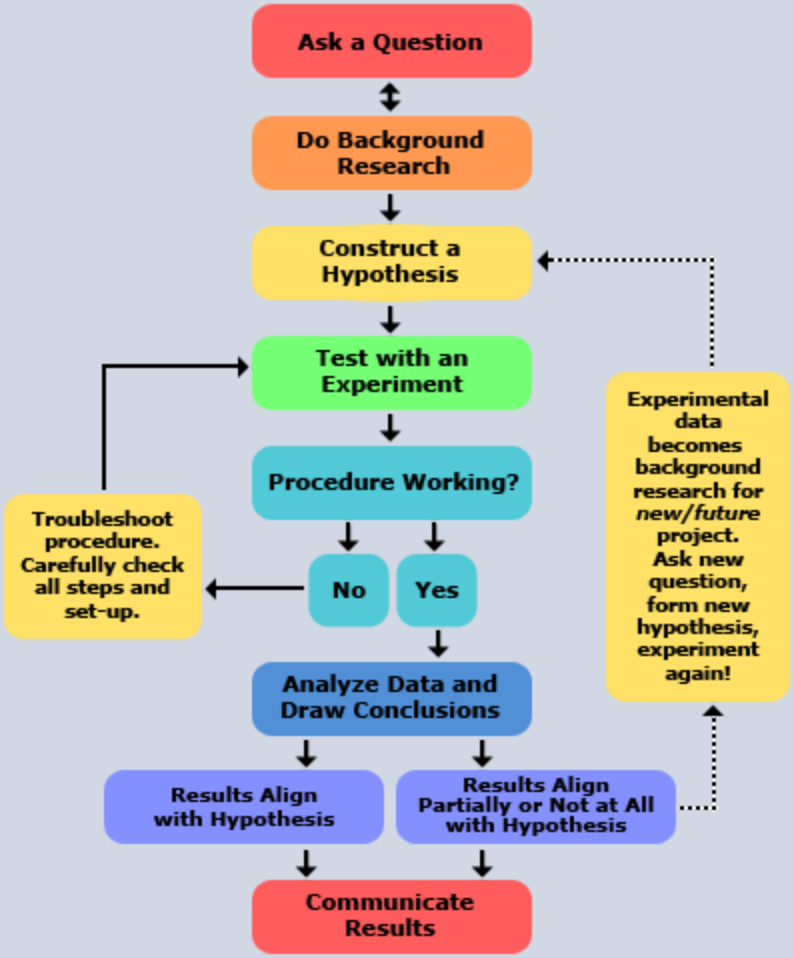
purpose of reviewing

- attempt to ensure that papers published in a reputable journal or conference are
    - trustworthy and
    - high-quality work
- Witing is not the end of the research process, but instead shapes it

skepticism (scepticism)

- A questioning attitude or doubt towards one or more putative instances of knowledge which are asserted to be mere belief or dogma
- A unifying principle for the scientific culture that determines the value of research
- A skeptic seeks the most accurate description or solution that fits the known facts
- without concern for issues such as the need to seek favour with authorities
- Most scientists test the reliability of certain kinds of claims by subjecting them to a systematic investigation using some type of the scientific method
- *pseudoscience*: Improperly apply or ignore the fundamental aspects of the scientific method
- Absolute skepticism is unsustainable
- *credulity* is pointless
- Impossible for knowledge to progress without some degree of questioning

Steps of the Scientific Method(I)



Steps of the Scientific Method(II)

1. Ask a Question
    - Your Question
    - Laboratory Notebook
2. Do Background Research
    - Background Research Plan
    - Finding Information
    - Bibliography
    - Research Paper
3. Construct a Hypothesis
    - Variables
    - Variables for Beginners
    - Hypothesis
4. Test Your Hypothesis by Doing an Experiment
    - Experimental Procedure
    - Materials List
    - Conducting an Experimen
5. Analyze Your Data and Draw a Conclusion
    - Data Analysis & Graphs
    - Conclusions
6. Communicate Your Results
    - Final Report
    - Abstract
    - Display Board
    - Science Fair Judging

Steps of the Scientific Method(III)

1. Formation of a precise question, the answer to which will satisfy the aim of the research
2. Development of a detailed understanding, through reading and critical analysis of scientific literature and other resources
3. Gathering of evidence that relates to the question, through experiment, analysis,or theory. These are intended to support—or disprove—the hypothesis underlying the question
4. Linking of the question and evidence with an argument, that is, a chain of reasoning
5. Description of the work in a publication.

End